

LYRA-CPP — THETIS DIRECT-PORT PLAN (TX)

Authoritative doc for the current TX work direction.

[!] The older EXECUTION_PLAN.md (Step 14 wire-layer rebuild, Phase 3 etc.) describes a DIFFERENT approach that is now superseded for the TX work. Do NOT mix the two plans — that doc tracks an earlier wire-rebuild pass which was partial and Lyra-native-leaning. THIS doc is the active plan going forward. Old doc remains in tree for historical reference only (some shipped pieces from it are still in production — noted per-component in the registry below).

Branch: tx-rebuild (the direct-port work lives here; origin/main carries the earlier Step 14 / TX-1 line and is deliberately untouched by the direct-port arc).

Reference tree (the ONLY trusted TX reference):

- D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\Project Files\Source\
 - wdsp*.c — WDSP DSP engine (GPL v3+, port-with-attribution OK)
 - ChannelMaster*.c — wire layer (GPL v3+, port-with-attribution OK)
 - Console*.cs — C# UI/control glue (study ONLY; write Lyra-native equivalents)

- Y:\Claude local_hl2src\ — HL2+ ak4951v4 gateway RTL

(control.v / radio.v / usopenhpsdr1.v etc. — the ground truth when host-side reference and gateway disagree)

Provenance & attribution rule (operator-locked 2026-06-09, supersedes the 2026-05-31 "no-attribution-in-commits" tightening):

Lyra-cpp is a derivative work of openHPSDR Thetis (ChannelMaster + WDSP modules), GPL v3+. The GitHub project description openly states this: **"TX baseline ported from openHPSDR Thetis (ChannelMaster) with Lyra-native DSP additions."** Open attribution is both legally required (GPL) and operator-intended (public project posture).

1. **License attribution at file head — REQUIRED.** Any file ported from WDSP or ChannelMaster carries a header block crediting upstream (NR0V for WDSP modules; openHPSDR Thetis for ChannelMaster modules) + the GPL v3+ notice. Existing AAMix.{h, cpp}, CMaster.cpp, RadioNet.cpp etc. already follow this; keep it.
2. **Code comments citing source — ENCOURAGED.** Citations may name the reference openly (Reference: Thetis ChannelMaster cmaster.c:297-313, // Per openHPSDR networkproto1.c:1078, // HL2+ ak4951v4 gateway control.v:209-220). The file:line is the primary technical content; the reference-app name is context that helps future readers find the source tree.
3. **Commit messages — REFERENCE-APP NAMES PERMITTED.** A commit like **"port aamix.c from Thetis ChannelMaster with Lyra-native idiom translations"** is fine — it matches the public GitHub posture and makes the history grep-able. The earlier 4cef88d tightening (2026-05-31) that put commit messages on the no-attribution list is **rescinded**.
4. ****Public docs (README, docs/*.md design docs, memory files, this plan) — OPEN ATTRIBUTION.**** Same posture as the GitHub description.
5. **User-facing UI strings — KEEP OPERATOR-FOCUSED.** UI labels operators see in normal operation (panel titles, button text, error toasts) describe Lyra behavior, not provenance. **"AAMix output (HL2 jack)"** not **"Thetis-porting AAMix output"**. About dialog / credits screen / install docs MAY (and probably should) name openHPSDR Thetis + WDSP/NR0V openly — credit where due. This is the only category that stays "Lyra-native voice in operator-facing labels," because operators interact with Lyra, not its source provenance.
6. **The "Lyra-Native style governs surrounding architecture only" rule (Stage B-locked) STANDS unchanged.** Reference DSP API call patterns are byte-for-byte faithful; Qt/Vulkan/QML/process model is Lyra-native.
7. **Reference precedence STANDS unchanged.** Verify TX ONLY against Thetis 2.10.3.13 (D:\sdrprojects\OpenHPSDR-Thetis-2.10.3.13\...) + the HL2+ ak4951v4 gateway RTL (Y:\Claude local_hl2src\). **Old**

Python Lyra is NOT a trusted TX reference (its TX never worked).

Consequence of the 2026-06-09 amendment: the three historical commits (efdde02, 604d87b, ed9f0cd) that "predated the no-attribution rule" and contained "Thetis" in their messages are **no longer in violation** — they're now consistent with the rule. Task #73 ("Cleanup pre-existing 'Thetis' mentions in shipped code") is **OBSOLETE**d by this amendment.

LOCKED METHODOLOGY (Stage B-validated, do NOT relax)

These rules were earned through the Stage B aamix.c port arc (2026-06-08) — see SESSION_LOG.md for the full trail.

1. ****Smallest revertable empirical step -> operator HL2 bench ->**

next step.** Each commit is one focused change with a falsifiable bench gate. Multi-step "design then implement" is for stages that genuinely need it (operator approves); default is iterate-and-bench.

2. **"Do as Thetis does, Lyra-Native style"** (locked

2026-05-30 after the fexchange0 crash arc, re-validated tonight after the B.6.b shortcut attempt cost a revert cycle). Three sub-rules:

- Every WDSP API call matches the reference byte-for-byte

(parameter values, order, the SET of setters fired, the lifecycle stage at which they fire). If tempted to pick a value the reference doesn't use, STOP and find what the reference uses. "I have a reason to pick differently" means I don't. Reference has been right every time.

- "Lyra-Native style" governs surrounding architecture ONLY

(Qt + Vulkan + QML + process/thread model + facade APIs + IPC + UI shell). The WDSP DSP-engine call surface is reference-bound.

- Provenance only in code COMMENTS + design docs + memory.

Never commit messages, never UI strings.

3. **Counters first when the symptom is "nothing happens".**

The Stage B 2-stage instrumentation (chain counters in xMixAudio/mix_main + dispatch counters in dispatchAudioFrame) localized the silent-audio defect to a single line of code on the 2nd bench. Pure speculation would have cost a multi-round revert/speculate cycle. Atomic counter + rate-limited qInfo = trivial overhead, very high diagnostic ROI when "I press play, hear nothing, log looks normal."

4. **Operator-empirical rule overrides agent inference.**

When operator hardware data contradicts what an audit/red-team/Plan-agent concluded, the data wins. Do not defend the audit — re-open, bisect, fix. This rule has fired repeatedly through the broader project arc; the Stage B B.6.b-fix1 fix was an instance (chain counters healthy suggested AAMix was correct; the temptation was to revert the whole port; the empirical pattern instead pointed at the wire-up surface where the bug actually was).

5. ****Read the reference FIRST, in shipped-code form, before**

designing.** Every time we designed first and read second, we shipped a regression. Every time we read first, we got the right answer in one pass. The no-attribution rule applies to shipped code; it does NOT mean "don't read the reference." Read it, dossier the file:line in the plan doc, then write Lyra-cpp.

6. **Session-start discipline:** git fetch origin is the

FIRST action every session, before reading any local state. If divergence exists, surface it to the operator BEFORE writing code. No exceptions.

7. **Push after every component ships, not at EOD.** Each

clean operator HL2 bench -> backup bundle + push immediately. Smaller in-flight delta = smaller surprise blast radius.

8. **Backup bundle naming:** _backups/lyra-cpp-YYYY-MM-DD-.bundle.

git bundle create ... --all (covers all refs).

STATUS REGISTRY — what's done, in-progress, pending

■ 2026-06-13 — TX BRING-UP COMPLETE (P0 → P4.b) + SHIPPED.

The verbatim direct-port wire chain is LIVE and operator-HL2-bench-passed:

SSB voice both sidebands, Phase-3-EXIT RF-safety kill-test, TUN

*Thetis-exact zero-beat, and **MSHV** → **TCI** → **FT8 real on-air QSOs***

(incl. F4FLF ~4160 mi DX). §7 dead-code retirements done. Released as

v0.2.3 (main == tx-rebuild == e8aafbc; tags v0.2.3 +

tx-working-2026-06-13; *GitHub Release + installer published*). Normal

feature dev resumes off main. Next milestones: PC/VAC mic-input paths

(#102/#103/#104), then v0.3 PureSignal (the P0.d _cmaster struct

already carries the full PS surface — no retrofit).

■ **2026-06-15 — VAC (#102/#158) SHIPPED as v0.3.0; post-P4 ordering LOCKED.**

The next-milestone PC/VAC mic-input path shipped: the ivac.c →

wire/IVAC device-layer rework (single full-duplex PortAudio stream,

*VAC/TCI crash fixes, RX-muted-on-TX) released as **v0.3.0** (main ==*

*tx-rebuild). **Open loose-end from that arc:** the TX MIC/ALC meters*

*were re-homed onto GetTXAMeter(chid(1,0)=1) *inside* the VAC work*

("#158 (post-DL) re-home" in metermodel.cpp / wdsp_engine.cpp) but

never got their own bench gate — operator reports MIC/ALC read nothing.

*Treat as **UNVERIFIED, not done** (the WDSP-TX-chain leveler/ALC rows are*

[WIP] "confirm reference-parity"). Closing it = a deep Thetis read of

the TX-meter read/display path (which txAMeterType, which channel,

where in the chain) + port + bench gate. Code-level contributors found

so far (candidates, NOT yet root-caused — do not patch blind): the

secondary-readout line is still hard-stubbed "-" (formatSecondaryText,

metermodel.cpp:577-582); and the AK4951 codec mic delivers zero into the

TXA until a rate change because the at-open seed (hl2_stream.cpp:837)

sets the rate bits but not mic_decimation_factor (defaults 0 = the

++count == factor harvest gate never fires).

>

LOCKED post-P4 feature ordering (operator decision 2026-06-15):

1. TX MIC/ALC metering — Thetis-reference read + port + bench

(closes out the VAC arc; verify-first, counters-first per the locked

methodology — do NOT patch the candidates above blind).

2. Stage F — TX DSP chain (EQ / compressor-Combinator / leveler /

ALC + their meters; broad TX value; feeds the in-flight #49 Profile

Manager).

3. RX2 + Split (#96–#101) — daily-use value, AND it builds the

(mox, ps_armed, rx2_enabled) DDC-dispatch state machine that PS

reuses (RX2 lives on DDC1; PS commandeers DDC0/DDC1 during MOX+PS).

4. PureSignal (Stage G/H, v0.3) — LAST. Narrowest audience (needs

the HL2 hardware mod + PS gateway), deepest / most cross-cutting surface; lands on top of the dispatch foundation RX2 builds, no wire retrofit (the P0.d _cmaster struct already carries the PS surface).

Rationale: PS gets cheaper + safer AFTER RX2, not merely lower-priority. The plan's F-before-G was version cadence (v0.2.x dynamics → v0.3 PS), NOT a hard compile dependency — so RX2 slots between F and PS.

■ **2026-06-17 — VAC closeout v0.3.1 + Stage-F native TX DSP rack + Profiles + TX monitor SHIPPED (post-P4). ** All native (Lyra-side, NOT direct ports — see the SHIPPED-WORK LOG entry for hashes). **v0.3.1** (a3a517f, 2026-06-15) closed the VAC arc: the enable/disable crash fix, Vol/Mute routed to VAC, AND the TX MIC/ALC/LEV metering the v0.3.0 banner above flagged UNVERIFIED is now wired Thetis-format (9594bd5 #160) — so that loose-end is **resolved in code** (the WDSP leveler/ALC rows below are flipped to [DONE] for their cffi bindings + metering wiring). NOTE: the #160 metering **bench close-out + Brent/Timmy v0.3.1 field-confirmation are still IN PROGRESS** — the meters are shipped, not yet field-validated.

Stage F TX DSP rack (native):

8-band parametric EQ (#50), Speech processor / Auto-AGC leveller + de-esser (#88), X-Air-style Combinator (#51), Plate (#52), with digital-mode auto-bypass + digital/CW lamp dim (#163). **Profiles #49** rack capture/apply (schema v3) + sideband/band-agnostic SSB profile + panel-lock fix + Speech-profile capture/lamp refresh (#162). **TX audio monitor #90** — 3 reference-faithful routes (HL2 jack / VAC device / TCI RX-audio) + the inline Audio-panel "Out" picker & MON→MON-TX relabel (#164). Released as **v0.3.1** (main == tx-rebuild); the native DSP/profile/monitor work rides on top (HEAD dc45485). **NEXT (current open queue — one task):** an AM/DSB RX-DSP bug — AM and DSB produce only the upper sideband (lower missing); the WDSP passband for AM/DSB is likely set one-sided instead of symmetric -W..+W. (GitHub / main-branch posture questions pending operator discussion.)

Format: per-file or per-component row. Status tags (ASCII so they render identically in the PDF/DOCX siblings):

- [DONE] DONE + operator-bench-validated
- [WIP] IN-PROGRESS (committed but not yet bench-validated, or partially shipped)
- [PEND] PENDING (planned, not started)
- [N/A] N/A (Lyra-native equivalent already shipped; no direct port planned — see notes column)

Wire layer (ChannelMaster — port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
WDSP exports linkage (openChannel/fexchange0/analyzers/PS family)	src/wire/wdspcalls.{h,cpp} (P0.a)	[DONE]	P0.a e0c0f4a 2026-06-09 . The single operator-approved linkage seam: fn-ptr table carrying the WDSP exports' exact names, resolved once from the loaded wdsp.dll. PureSignal entry family complete (verified vs dumpbin). RESAMPLE-typed + flush_resample added at P0.c.
ChannelMaster\cmcomm.h (family common header)	src/wire/cmcomm.{h,cpp} (P0.b -> P0.d)	[DONE]	complex typedef + PORT + verbatim malloc0 (wdsp/utilities.c:37-43) + PI + the reference include surface. P0.d added the opaque verbatim twin typedefs for deferred-subsystem types (ANB/NOB/VOX/TXGAIN/ANALYZERS + DELAY at P2.a); EER completed at P2.a 2026-06-12 — full verbatim struct (wdsp/eer.h:30-49) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under #pragma once).
ChannelMaster\ilv.c (TX I/Q interleaver)	src/wire/ILV.{h,cpp} (P0.b)	[DONE]	P0.b 1f49d98 2026-06-11 . VERBATIM direct port (ilv, *ILV twin typedef, raw Outbound fn ptr, pcm->xmtr[].pilv setters, malloc0/_aligned_free). Mechanical diff vs ilv.{h,c} = whitespace-only. The earlier src/wdsp/ILV retrofit + its pilv[] bank are retired. scratch/test_ilv ALL PASS.
ChannelMaster\cmbuffs.{h,c} (CMB ring + cm_main pump)	src/wire/CmBuffs.{h,cpp} (P0.d)	[DONE]	P0.d afc7950 2026-06-12 . VERBATIM rewrite (cmb, *CMB, #define CMB_MULT (3), malloc0/_aligned_free; the Phase-C retrofit's calloc/intptr_t/guard deviations removed; pcm->in[] alloc moved back to create_cmastep per reference). Mechanical diff = whitespace-only. [DONE] P0.d operator HL2 RX-regression bench PASSED 2026-06-12 .
ChannelMaster\cmsetup.{h,c} (radio structure + id helpers)	src/wire/cmsetup.{h,cpp} (P0.d, NEW)	[DONE]	P0.d afc7950. VERBATIM: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. main.cpp config: SetRadioStructure(2,1,1,1,0,...) -> chid(0,0)=0 RX1 / chid(1,0)=1 TX matches the live channel layout. [DONE] Bench PASSED 2026-06-12.

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\cmaster.{h,c} (FULL: struct + create/destroy + xcmaster + setters)	src/wire/CMaster.{h,cpp} (P0.d)	[DONE]	P0.d afc7950. VERBATIM rewrite: full _cmaster struct (cmaster.h:39-99 incl. the PS surface — out[3]/panalalloc/pgain/peer), cmaster, *CMaster, raw TCI callback fn ptrs, enum AudioCODEC, cm = {0}. Verbatim no-arg create_xmtr/destroy_xmtr (out[0..2] malloc0 + OpenChannel(ch 1) + XCreateAnalyzer(TX disp 1) + create_ilv through the wdspcalls seam — the TxChannel RAIL carve-out is DELETED , src/wdsp/TxChannel.{h,cpp} retired). create_cmater/destroy_cmater verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams). xcmaster verbatim (update[] CS restored, real stype/txid/chid, TCI memset restored; fexchange0 + monitor xMixAudio + xilv live). All Sendp*/Set* setters ported. Deferred subsystem lines (dexp/anti-vox/txgain/sidetone/pipe/cmasio/analyzers.c/create_rcvr-body) carried IN PLACE as reference text with DEFERRED tags; eer RESTORED VERBATIM at P2.a 2026-06-12 (create_eer run=0 / destroy_eer / xeer / pSetEER* via the wdspcalls seam — the object exists so the sendOutbound/sendProtocol1Samples peer->run derefs are valid, per cmaster.c:212-224). Sole code accommodation: (char*)" at XCreateAnalyzer. [DONE] Operator HL2 RX-regression bench PASSED 2026-06-12 (RX working on the new per-stream pump/ring layout).
ChannelMaster\obbuffs.c (TX-out ring/seam)	src/wire/ObBufs.{h,cpp} (P1, NEW)	[DONE]	P1 75754c2 2026-06-12 — obbuffs.{h,c} verbatim (obb/*OBB twin typedef, file-scope obp four-alias bank, the reference's own calloc/free 2014-era idioms kept). Mechanical diff: .cpp IDENTICAL; .h sole delta = the include-guard define replaced by #pragma once (documented packaging). The sendOutbound(id, a->out) ob_main hand-off was restored at P2.c (no longer deferred). Wire-LIVE since P4.b 2026-06-13 — create_obbuffs/ob_main are live in the direct-port TX chain (TU no longer dormant); was [WIP] SHIPPED while dormant, now DONE in the live chain.
ChannelMaster\networkproto1.c (HL2 wire writer)	src/wire/NetworkProto1.c pp (sendProtocol1Samples) + FrameComposer write_main_loop_hl2 (LIVE)	[DONE]	Wire-LIVE at P4.b 2026-06-13. sendProtocol1Samples verbatim drives the EP2 writer thread; FrameComposer monolithic write_main_loop_hl2 is the WriteMainLoop_HL2 equivalent (#121/#122 fold). The old hl2_stream.cpp Step-14 mix retired at §7.
ChannelMaster\network.c (UDP + keeplive)	src/wire/ObBufs.{h,cpp} ob_main + sendOutbound (LIVE)	[DONE]	Wire-LIVE at P4.b. sendOutbound verbatim (P2.c); the Step-14 Lyra-native OutboundRing + Ep2SendThread translations RETIRED at §7 (fe6a438).

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\netInterface.c: :create_rnet	src/wire/RadioNet.cpp::c reate_rnet (LIVE)	[DO NE]	Direct port shipped earlier. The Stage-A SendpOutboundRx no-op stub at line 272 was DELETED 2026-06-08 (B.6.b-fix1, commit 8b8e0da) because it clobbered the AAMix wire-up; future codec-id switching must NOT re-introduce a default registration here. P3 = the outbound registrations (SendpOutboundTx(OutBound) etc.) — MUST register AFTER create_xmtr (the verbatim setters have NO null guards, reference ordering).

WDSP RX chain (port directly w/ attribution)

Source	Target (lyra-cpp)	Status	Notes
ChannelMaster\aamix.{c,h}	src/wire/AAMix.{h,cpp} (P0.c)	[DONE]	P0.c e0f2584 2026-06-11 — VERBATIM re-port (supersedes the 2026-06-08 Stage-B C++23-idiom translation, which the operator rejected as a deviation class). aamix, *AAMIX twin typedef + anonymous slew struct, raw Outbound fn ptr, paamix[] bank private to the .cpp, Win32 primitives direct. Mechanical diff vs aamix.{h,c} = whitespace-only. src/wire/resample.h = the public RESAMPLE ABI verbatim. WdspEngine RX wire-up: capturing lambda -> static member fn + TU-scope context ptr (the reference free-fn+global shape). [DONE] Operator HL2 RX bench PASSED 2026-06-11.
wdsp\firmin.c (filter cores)	(consumed via WDSP cffi)	[N/A]	Used internally by WDSP RXA chain via cdef + dlsym. No source-level port needed (the DLL ships with Lyra-cpp via _native/ per the Python-project pattern; cdef + dlsym in wdsp_native.cpp). WISDOM cache fix shipped on Python side; lyra-cpp's RX chain inherits the same path.
wdsp\wisdom.c	(consumed via WDSP cffi)	[N/A]	Same as firmin.c.

WDSP TX chain (per CLAUDE.md §4.1 port table)

Source	Target (lyra-cpp)	Status	Notes
wdsp\TXA.c (channel scaffold)	(consumed via wire/wdspcalls seam — P0.a/P0.d)	[DONE]	TxChannel class DELETED at P0.d afc7950 — the verbatim create_xmtr in wire/CMaster.cpp opens the WDSP TXA channel itself (OpenChannel/fexchange0/CloseChannel through the wdspcalls seam, cmaster.c:177-190 byte-exact), which removed the runtime-DLL RAIL justification the class existed for. wdsp_native.{h,cpp} still carries the SetTXA*/GetTXAMeter cdefs for the operator-control surface.
wdsp\bandpass.c (bp0/bp1)	(consumed via WDSP cffi — SetTXABandpassFreqs)	[N/A]	The §15.23 trap (SetTXABandpassRun overrides bp1 to a stale state) is documented in CLAUDE.md + project_lyra_cpp_tx.md. cffi-only access is the correct posture.
wdsp\compress.c (TX compressor)	src/wdsp/Compress.{h,cpp} OR cffi binding	[N/A]	Superseded by the native TX DSP rack (post-P4). Operator chose the Lyra-native path: the Speech processor (#88, 902b814/93adb39) + Combinator (#51) cover speech dynamics natively, pre-WDSP-TXA. No WDSP compress.c port/cffi binding shipped.
wdsp\cfcomp.c (5-band CFC)	src/wdsp/CFComp.{h,cpp} OR cffi binding	[N/A]	Superseded by the native Combinator #51 (013bf12/61e16c4/b53ae35) per the §15.19 plan — Lyra-native multiband compressor replaces the WDSP CFC. No CFC port shipped.
wdsp\osctrl.c (CESSB)	cffi binding (SetTXAosctrlRun)	[PEND]	Likely cffi-only; ~200 LOC port not justified unless operator UI need surfaces.
wdsp\wcpagc.c mode 5 (leveler)	cffi binding (SetTXALeveler*)	[DONE]	cffi binding live in the wire-LIVE TXA chain. TX LEV metering wired Thetis-format at #160 9594bd5 (v0.3.1) + #49 profile Leveler fields — addresses the v0.3.0 "UNVERIFIED" metering loose-end (shipped in code; #160 bench close-out / tester confirm still IN PROGRESS).
wdsp\wcpagc.c mode 5 (ALC)	cffi binding (SetTXAALCMaxGain)	[DONE]	cffi present; SetTXAALCThresh does NOT exist (§15.23-class trap — only MaxGain governs ALC ceiling). TX ALC metering wired Thetis-format at #160 9594bd5 (v0.3.1) — addresses the v0.3.0 "UNVERIFIED" flag (shipped in code; #160 bench close-out / tester confirm still IN PROGRESS).

Source	Target (lyra-cpp)	Status	Notes
wdsp\eqp.c (parametric EQ)	src/wdsp/EQ.{h,cpp} OR cffi binding	[N/A]	Superseded by the native EQ #50 (edbaa52/17a28b7/eb92231) — native RBJ-biquad 8-band parametric EQ (ParamEq engine + EESDR3-style panel), wired pre-WDSP-TXA. No WDSP eqp.c port shipped.
wdsp\gen.c (gen0/gen1 generators)	cffi binding (SetTXAPreGen*/SetTXAPostGen*)	[WIP]	gen1 (postgen) live for TUN tone — wire-LIVE + bench-passed since P4.b 2026-06-13 (Thetis-exact zero-beat). gen0 (pregen) cdefs still deferred per §15.23 (bench-tooling nicety, not on the signal path) — hence row stays [WIP].

TX PureSignal (per CLAUDE.md §4.1 v0.3)

Source	Target (lyra-cpp)	Status	Notes
wdsp\iqc.{c,h}	src/wdsp/Iqc.{h,cpp} (~315 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 5-state PS lifecycle (RUN, BEGIN, SWAP, END, DONE).
wdsp\calcc.{c,h}	src/wdsp/Calcc.{h,cpp} (~1164 LOC)	[PEND]	v0.3. cffi math + Lyra-cpp wrapper for the 8-state PS FSM + 3-state auto-attenuator FSM + PSDialog UI.
wdsp\lmath.c::xbuilder	src/wdsp/PsXBuilder.{h,cpp} (~200 LOC)	[PEND]	v0.3. Cubic-spline coefficient builder, used by Lyra-cpp-side calcc orchestration.
wdsp\delay.c	src/wdsp/DelayLine.{h,cpp} (~80 LOC)	[PEND]	v0.3. TX/feedback time-alignment.

Console (C# — study only, Lyra-native equivalents)

Source	Target (lyra-cpp)	Status	Notes
Console\console.cs::chkMOX_CheckedChanged2	src/ptt.{h,cpp} + src/wdsp_engine.cpp keydown/keyup	[DONE]	Lyra-native FSM. Verified against the reference's keydown/keyup ordering invariants (RX-DSP stop on keydown -> wire MOX -> TX-DSP start; keyup TX-DSP off -> mox_delay -> clear MOX -> ptt_out_delay -> RX-DSP restart). Live + operator-HL2-bench-passed at P4.b 2026-06-13 (SSB voice both sidebands + RF-safety kill-test exercise the FSM end-to-end). No direct port — C# is study-only by license.
Console\console.cs (TR delays)	src/tx/TrSequencing.{h,cpp} (LIVE)	[DONE]	Lyra-native shipped with operator's Thetis-DB-verified defaults (mox=15/rf=50/sp ace_mox=13/ptt_out=5/key_up=10 ms).

Source	Target (lyra-cpp)	Status	Notes
Console\console.cs::m_bATTonTX	src/hl2_stream.cpp fsmAdvance + compose_case_11 (LIVE wire)	[WIP]	Lyra-native. Reference posture: keydown force step-att 31 (PS-A off OR CW); keyup restore. WIRE path traced live + functional 2026-06-17 : fsmAdvance raises setTxStepAttnDb(31) on keydown (hl2_stream.cpp:1651) → tx_step_attn = 31-31 = 0 → XmitBit-gated compose_case_11 C4 = 0x40 → ak4951v4 cmd_addr 0x0a rx_gain = 0 = min LNA during TX. BUT NOT [DONE] : (a) the src/tx/AttOnTxPolicy.{h,cpp} class is unused / wire-inert (its own header says so) — the live mechanism is inline in hl2_stream.cpp, the earlier doc citation was wrong; (b) ~~operator-reported 2026-06-17: NO UI surface~~ → UI surface BUILT 2026-06-17 (\$15.31) : TxPanel "ATT" lamp in the Mic↔Tune gap (gray ATT off / orange ATT 31 armed-RX / solid-red ATT -31 engaged-TX, AUTO/TUN/MOX idiom) + Settings → TX → "ATT on TX (RX-ADC protection)" group (Enable + dB 0..31 spin); operator-gated, default ON/31, QSettings tx/attOnTx*; fsmAdvance now gates on the toggle + uses the value (replaces the hardcoded kAttOnTxDb). Operator-approved in review; awaiting on-air bench confirm to flip [DONE] . The AttOnTxPolicy class stays unused (flag: delete-or-wire-up later). Broader Task #114 still PENDING (panadapter TX offset + PA-enable safety).
Console\PSForm.cs	src/ui/PSDialog.{h,cpp} (NEW, v0.3)	[PEN D]	v0.3 PureSignal UI. C# study only; write Qt/QML native.
Console\HPSDR\IoBoardHl2.cs	src/hl2_stream.cpp (LIVE)	[DON E]	HL2 I/O quirks implemented Lyra-native via the protocol-byte verification + gateway-RTL ground-truth pass (CLAUDE.md §15.26). Live + bench-passed across the RX + P4.b TX chain (incl. on-air FT8).

STAGE INDEX — ordered work plan

P0 verbatim-rewrite arc (2026-06-09 -> current; the ACTIVE plan)

After the operator rejected the 2026-06-09 "rule-#8" retrofit deviations ("I said PORT!"), every TX-porting file is being rewritten VERBATIM (byte-faithful; only documented packaging differences — lyra::wire namespace, PORT-expands-empty, the wdspcalls seam for the statically-linked WDSP exports). Each step ships with a mechanical diff vs the reference + an operator HL2 bench gate.

Step	Status	What	Commit / gate
P0.a	[DONE] DONE	wire/wdspcalls.{h,cpp} — the single approved WDSP linkage seam (exact export names, PS entry family complete)	e0c0f4a 2026-06-09
P0.b	[DONE] DONE	ilv.{h,c} verbatim -> wire/ILV.{h,cpp} + new wire/cmcomm.{h,cpp}	1f49d98 2026-06-11; test_ilv ALL PASS

Step	Status	What	Commit / gate
P0.c	[DONE] DONE	aamix.{h,c} verbatim -> wire/AAMix.{h,cpp} + wire/resample.h RESAMPLE ABI	e0f2584; [DONE] operator HL2 RX bench PASSED 2026-06-11
P0.d	[DONE] DONE	cm buffs/cm master/cm setup verbatim -> wire/CmBuffs/wire/CMaster/wire/cmsetup; full _cmaster struct (PS surface); TxChannel carve-out DELETED	afc7950 2026-06-12; [DONE] operator HL2 RX-regression bench PASSED 2026-06-12
P1	[DONE]	ob buffs.c verbatim -> wire/ObBuffs.{h,cpp} (the TX-out seam; separate TU from cm buffs; sendOutbound call restored at P2.c)	75754c2 2026-06-12; shipped dormant, then wire-LIVE at P4.b 2026-06-13 (create_ob buffs/ob_main live in the direct-port TX chain); clean build 0 warnings; mechanical diff IDENTICAL
P2	[DONE]	sendOutbound / sendProtocol1Samples fidelity audit of the dormant wire layer — COMPLETE 2026-06-12 (P2.a eer completion + P2.b prn outbound surface verbatim + P2.c wire/Network.cpp sendOutbound verbatim, mechanical diff IDENTICAL; the ObBuffs ob_main hand-off restored). P2.a SHIPPED 2026-06-12 — eer completion (verbatim struct + seam entries + the four CMaster restore sites); kills the null-peer landmine both reference functions deref. NEXT: P2.b prn outbound surface verbatim (outLRbufp/outlQbufp/OutBufp -> reference pointer fields + the HANDLE semaphore quartet; callers bend) -> P2.c wire/Network.cpp sendOutbound verbatim (P1 branch live, ETH branch DEFERRED) + restore the ObBuffs ob_main call.	Audit finding: Lyra's Step-14 OutboundRing/Ep2SendThread are functionally-parallel idiom translations (vector buffers, bool+cv semaphores) predating the verbatim mandate — retired at P4 when the verbatim chain goes live.
P3	[DONE]	netInterface registrations + ob buffs ring lifecycle. SHIPPED + operator HL2 bench PASSED 2026-06-12 (RX fine; stop/start + shutdown clean) — create_rnet moved to once-per-process at the main.cpp QTimer AFTER create_xmtr (the reference C#-init order; fixes the per-open prn re-allocation leak); SendpOutboundTx(OutBound) restored at the reference site (netInterface.c:1761, tail of create_rnet); UpdateRadioProtocolSampleSize verbatim (netInterface.c:1836-1858, diff IDENTICAL) called per-open per StartAudio:45 — creates ob buffs rings 0/1 (2 new ob_main pump threads per session, idle: no producers until P4); destroy_ob buffs(0/1) at close() per StopAudio:112-113. RX side deliberately NOT registered (WdspEngine hybrid owns RX dispatch until P4 — the B.6.b clobber class).	[DONE] Operator HL2 bench PASSED 2026-06-12.

Step	Status	What	Commit / gate
P4	[DONE]	Wire-LIVE switchover — COMPLETE + operator HL2 bench PASSED 2026-06-13 . P4.a prep (2 wire-inert commits): sendProtocol1Samples verbatim in src/wire/NetworkProto1.cpp + io_keep_running + FrameComposer tail flipped to ReleaseSemaphore(hobbufsRun) + SetTXAPostGen wdspcalls + TxReadBufp double* / hWriteThreadMain HANDLE. P4.b SHIPPED in 3 commits: fb9ec41 Wire-LIVE RX-out gate, 0b551b4 first modulated RF through the direct-port chain, 9db2c50 SSB sideband fix (USB transmits USB, 9100-verified). Plus 84dbb12 TUN panadapter-spike fix (Thetis-exact zero-beat confirmed, display-only bug). §7 retirements DONE (fe6a438 OutboundRing+Ep2SendThread, 99fd24a txWorkerLoop+composeCC+keepalive, e711256 old HL2-jack EP2 ring). TCI digital TX re-home bd61d07 (TciTxBridge fills the §10.3 skeleton; on-air FT8 QSOs). Released v0.2.3.	Bench gates PASSED: SSB voice both sidebands + RF-safety kill-test + MSHV→TCI→FT8 on-air.

PureSignal is a committed feature — the P0.d _cmaster struct carries the full PS surface so v0.3 lands on top without a retrofit.

Historical stage index (pre-P0 lettered stages — superseded)

The original lettered plan below is retained for history. The P0 arc absorbed/superseded it: Stage C (ilv) = P0.b; Stage D (xcmaster) = P0.d; Stage E (TxChannel audit) = obsoleted by the P0.d TxChannel deletion; Stage B's idiom-translated AAMix was re-reported verbatim at P0.c.

Stage	Status	What	Notes
A	[DONE] DONE	CMaster.cpp wire-up stubs (Send*Outbound* + SetTCIRun)	Superseded by the P0.d full verbatim CMaster port.
B	[DONE] DONE	aamix.c direct port + wire into RX path	Shipped 2026-06-08 (idiom-translated); re-reported VERBATIM at P0.c.
C	[DONE] DONE (as P0.b)	ilv.c direct port (TX I/Q interleaver)	Landed verbatim at wire/ILV.{h,cpp}.
D	[DONE] DONE (as P0.d)	xcmaster pump body direct port	Landed verbatim in wire/CMaster.cpp (real stype/txid/chid + update[] CS).
E	[N/A] OBSOLETE	TxChannel reference-parity audit + reconciliation	TxChannel DELETED at P0.d — the verbatim create_xmtr replaces it.
F	[PEND] TBD	WDSP TXA chain components per CLAUDE.md §4.1 v0.2.1	compress / cfcomp / eqp / wcpagc — operator-driven per-feature direct port vs cffi-only call. Order TBD.

Stage	Status	What	Notes
G	[PEND] TBD	PureSignal direct ports (v0.3)	iqc / calcc / xbuilder / delay. Lands when v0.3 PS work starts. Needs operator HL2+ with the PS hardware mod.
H	[PEND] TBD	TX/PS auto-attenuator FSM port	C# study only (PSForm.cs); Lyra-native rewrite.

Ordering note: the P0->P4 sequence above is the active linear plan. F–H land after P4 Wire-LIVE. The methodology (smallest revertable step + mechanical diff + operator bench gate) applies within each step.

OPEN QUESTIONS FOR OPERATOR

These are decisions that shape the plan but I cannot make unilaterally — surface them when picking up tomorrow:

1. ****Stage C (ilv.c) vs Stage D (xcmaster) vs Stage E**

(TxChannel audit) — which first? All three are defensible:

- Stage C unlocks reference-faithful TX I/Q flow.
- Stage D removes the Lyra-native pump (cleaner end-state).
- Stage E formalizes what's already shipped (lowest risk,

highest confidence-gain).

2. ****Do we re-port networkproto1.c + network.c + the Step 14**

wire pieces direct from reference, OR leave them as the current mix of Lyra-native rewrites + original `hl2_stream.cpp`? The Step 14 plan was midway through these when Stage B started. Direct-port would replace the rewrites with reference-verbatim ports. Operator call.

3. ****WDSP TXA-chain components — direct port (~150-600 LOC**

each) vs cffi-only? The DLL exports work; cffi is faster to ship. Direct port gives Lyra-cpp full source visibility + lets us patch behavior at the source level. Per-component operator decision; default = cffi-only unless a real source-level need surfaces.

4. ****PS work timing — start v0.3 (iqc/calcc/etc.) now in**

parallel with TX, or sequence after TX is fully done? The CLAUDE.md plan defers PS to v0.3 (after v0.2 TX release). Operator may want to start the ports early.

5. ****The §15.27 "opt-in toggles that become defaults" cleanup**

from the Python project — does it have a lyra-cpp analog here? The Python project had a long backlog of legacy toggles to retire. Lyra-cpp doesn't carry the same legacy, but new direct-port commits may obsolete some Settings surfaces operator-knows-about — track per-component.

SHIPPED-WORK LOG

Newest at top. Cross-references SESSION_LOG.md for full arc details.

2026-06-14 -> 2026-06-17 — post-P4: VAC v0.3.0/v0.3.1, native TX DSP rack, Profiles, TX monitor (#90), Out picker

These are post-direct-port work: native (Lyra-side) TX-DSP / host-audio / profile features that land ON TOP of the verbatim P0–P4 chain. They are NOT direct ports (no Thetis ChannelMaster/ WDSP source counterpart) — logged here as a post-P4 section so the P0–P4 port-stage record above stays intact.

• **VAC / IVAC (#102/#158)** — PC/VAC mic-input + RX-to-PC path.

`ivac.c` -> wire/Ivac direct port (Stage 0/1 4831db7/52adff3),

#158 complete (RX decode + TX on-air + auto-start + profile VAC source) 6c3fecb. Released **v0.2.4** "Profiles + VAC" c48c94d; VAC mic mono-combine fix b57bd09 -> **v0.2.5** f55da2c. Device-layer rework DL-0..DL-5 (PortAudio 19.7.0 vendored static, single

full-duplex stream, Thetis-faithful host-API/device pickers, RX-muted-on-TX via SetIVACmox, old Qt two-stream ivac_audio retired): 11812f9..e8d0239; plus two crash fixes — VAC-teardown UAF f0c6497, TCI dangling-owner d508b8e. TX MIC/ALC/LVL meters re-homed onto wire-live GetTXAMeter 993e10e. Released **v0.3.0** accee55. Then **v0.3.1** a3a517f: VAC enable/disable crash fix + Vol/Mute routed to VAC 77dc55d (#161) + Thetis-format TX MIC/ALC/LEV metering + #49 profile Leveler/ALC fields 9594bd5 (#160). v0.3.1 addresses the v0.3.0-banner "TX metering UNVERIFIED" loose-end in code (#160 bench close-out / Brent+Timmy field-confirm still in progress). Interim VAC-UI fixes (88bdab1 relabel, b57bd09 combine, a01d308 output-"(none)") shipped along the way.

• **Stage F — native TX DSP rack (pre-WDSP-TXA mic rack):**

- **EQ #50** — native ParamEq (RBJ biquad cascade) edbaa52;

EqModel + EESDR3-style panel 17a28b7; wired into the live TX mic rack eb92231.

- **Digital-mode auto-bypass** of the native mic DSP (DIGU/DIGL)

c8441a6.

• **Speech #88** — native SpeechProcessor (Auto-AGC leveller + de-esser) 902b814; Speech panel + EQ analyzer + Reset-All 93adb39.

• **Combinator #51** — native engine 013bf12; model+panel+dock 61e16c4; wired into the live rack after EQ b53ae35.

- **Plate #52** — native Schroeder–Moorer engine ae9c5d3;

PlateModel + "Plating" panel 0cfd7e5; wired after Combinator e96a6e8.

- Rack-lamp dim in digital/CW + EQ analyzer recolor (#163)

9be1fc1 / 558d5d9.

• **Profiles #49 (TX/RX profile manager):** rack

saveState/loadState round-trip e0f264e (Stage 1); schema v3 bundling the native rack blobs b453320 (Stage 2); rack wired into profile capture/apply 29f9b92 (Stage 3). Sideband- and band-agnostic SSB profile 296cafc. Panel-lock float-drag fix a82eb77 (exclude TX rack panels from LOCK) + lock also blocks resize/custom-float cf976c1. TX Speech profile capture + toggle-lamp refresh on recall (#162) c17960c / 46a428f.

- **TX audio monitor #90** — Route 1 (MON button + Monitor slider,

HL2 jack) 6e6a282; Routes 2+3 (VAC device + TCI RX-audio) e1fd186. Reference-verified against Thetis audio.cs.

- **#164** — inline Audio-panel "Out" output-device picker wired +

MON->MON-TX relabel 740f769.

- **USER_GUIDE** catch-up commits (minor): 66adc5e, 266d8cd,

dc45485.

2026-06-13 — P4 Wire-LIVE switchover + v0.2.3 release ([DONE] HL2 bench PASSED)

- **P4.a prep** (wire-inert): sendProtocol1Samples verbatim

a4aa8c3 + SetTXAPostGen/TxReadBufp/hWriteThreadMain field types baef866.

- **P4.b SHIPPED** (3 commits, operator HL2 bench PASSED 2026-06-13):

fb9ec41 Wire-LIVE RX-out gate, 0b551b4 first modulated RF through the direct-port chain, 9db2c50 SSB sideband fix (USB transmits USB, 9100-verified). Plus 84dbb12 TUN panadapter-spike fix (display-only, Thetis-exact zero-beat).

- **\$7 retirements DONE:** fe6a438 OutboundRing+Ep2SendThread,

99fd24a txWorkerLoop+composeCC+keepalive, e711256 old HL2-jack EP2 ring.

- **TCI digital TX re-home** bd61d07 (first FT8 QSOs via MSHV/TCI through the verbatim chain) + space-bar PTT toggle 2824ebf (#157).

- Released **v0.2.3** "Transmit (SSB + digital TCI) on air"

e8aafbc (tags v0.2.3 + tx-working-2026-06-13).

2026-06-12 — P0.d cmbuffs/cmaste/cmsetup VERBATIM port ([DONE] HL2 bench PASSED)

- **P0.d** afc7950 on tx-rebuild: NEW wire/cmsetup.{h,cpp}

(cmMAX* 16/4/4/2/32 + the 8 id helpers + SetRadioStructure + set_cmdefault_rates); wire/CmBuffs.{h,cpp} rewritten verbatim (cmb,*CMB,malloc0/_aligned_free; Phase-C retrofit deviations removed); wire/CMaster.{h,cpp} rewritten verbatim (FULL _cmaster struct incl. the PureSignal surface, enum AudioCODEC, raw TCI fn ptrs, verbatim create/destroy_cmaste + create/destroy_xmtr + xcmaster + all Sendp*/Set* setters).

- **TxChannel RAIL carve-out DELETED** (src/wdsp/TxChannel.{h,cpp}

retired) — the verbatim create_xmtr opens the WDSP TXA channel + TX analyzer (disp 1) + out[0..2] + ILV itself through the wdspcalls seam.

- main.cpp: SetRadioStructure(2,1,1,1,0,...) config before

create_cmaste (chid(0,0)=0 RX1 / chid(1,0)=1 TX = the live layout); create_xmtr() post-resolve_wdsp_calls (documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy_xmtr().

- Verification: clean build ZERO warnings; test_ilv ALL PASS;

mechanical diffs IDENTICAL (CmBuffs.{h,cpp} / cmsetup.h / CMaste.h / all 10 fully-verbatim cmaste.c functions) + live-line-subset PASS for the partially-deferred bodies (sole accommodation: (char *)"" at XCreateAnalyzer).

- New at startup: TWO cm_main pump threads (streams 0+1; stream 0

idles — no Inbound producer); TX stays wire-quiescent (OutboundTx null until P3).

- **[DONE] GATE PASSED 2026-06-12: operator HL2 RX-regression bench**

("RX still working") — P1 (obbuffs.c) is unblocked.

2026-06-09 -> 2026-06-11 — P0.a/P0.b/P0.c verbatim-rewrite arc

- Operator rejected the 2026-06-09 "rule-#8" retrofit deviations

("I said PORT!") -> every ported file rewritten VERBATIM.

- **P0.a** e0c0f4a: wire/wdspcalls.{h,cpp} — the single

approved WDSP linkage seam (fn-ptr table, exact export names, PureSignal entry family complete + dumpbin-verified). scratch/_typedef_fidelity_test.cpp disproved the claimed "C++ collisions" (twin typedef / complex[2] / raw fn ptr all compile verbatim).

- **P0.b** 1f49d98: ilv.{h,c} verbatim -> wire/ILV.{h,cpp};

new wire/cmcomm.{h,cpp} (complex/PORT/malloc0/PI); pilv[] bank gone; SendpOutboundTx raw fn ptr (register AFTER create_xmtr — no null guards, reference ordering). Mechanical diff whitespace-only; test_ilv ALL PASS.

- **P0.c** e0f2584: aamix.{h,c} verbatim -> wire/AAMix.{h,cpp}

(supersedes the Stage-B idiom translation); wire/resample.h = public RESAMPLE ABI; wdspcalls retyped + flush_resample; WdspEngine outbound lambda -> static fn + TU-scope context ptr. **[DONE] Operator HL2 RX bench PASSED 2026-06-11.**

2026-06-08 EOD — Stage B aamix.c port COMPLETE

- **Stage B (aamix.c)** [DONE] direct port shipped + bench-validated

- Commits 27ac2fa -> 7a50b07 on tx-rebuild

- Operator HL2+ bench 19:27:49 -> 19:28:47: audible RX through

both AK4951 jack and PC-Soundcard sinks; mute/unmute via dispatch; output-device swap; AGC mode changes; chain out=N tracks 1:1 with dispatch calls=N across both sinks

- B.6.b root cause: Stage-A SendpOutboundRx no-op stub at

RadioNet.cpp:272 clobbered aaMix_->Outbound ~1 ms after WdspEngine::openRx1 wired the real lambda. Fixed by deleting the stub (commit 8b8e0da).

- 2-stage diagnostic instrumentation (chain @ 225518c +

dispatch @ 12369bc) localized the defect to a single line of code on the 2nd bench — pure-speculation cost avoided.

- Methodology lessons locked into this doc § "LOCKED

METHODOLOGY" above (counters-first when symptom is "nothing happens"; operator-empirical rule held; "do as Thetis does" continues to pay).

- Tasks #130 (parent) + #132 (Stage B.6.b) closed.
- Backup _backups/lyra-cpp-2026-06-08-aamix-stage-B-COMplete-with-docs.bundle.

Earlier work

See SESSION_LOG.md for the full session-by-session log. The Step 14 wire-layer arc (Stages 1, 1.5, 2a, 2b1, 2b2) is covered in detail there + in EXECUTION_PLAN.md's progress- tracking sections. TX-1 components 2-6 (main branch) are covered in project_lyra_cpp_tx.md (memory file).

RESUME POINTER (next session)

First actions (locked discipline):

1. git fetch origin
2. git status + git log --oneline -5 tx-rebuild origin/main
3. Read this doc + the "READ FIRST" block in project_lyra_cpp_tx.md
4. Read SESSION_LOG.md newest entry for arc detail

Branch posture (RESOLVED 2026-06-17): main is the working trunk again — main == origin/main == tx-rebuild == the **v0.4.0** release commit. Operator consolidated onto main (TX largely done) to stop forgetting to fast-forward it. Do future work on main; tx-rebuild is a legacy ref.

SHIPPED since the last revision of this pointer: the AM/DSB one-sided bug is **FIXED** (it was the SSB-only TX modulator, not an RX-DSP bug — RX was correct; the one-sided signal was the operator's own transmit) →

native AM/DSB/SAM/FM transmit + AM carrier level shipped (#106/#93). Plus operator-visible ATT-on-TX (§15.31) and a UI-readability batch.

Released v0.4.0 (33 commits since v0.3.1; installer published to GitHub).

Next major candidate — CW transmit (#105): DESIGN MAPPED 2026-06-17. See docs/architecture/cw_tx_design.md (full port plan from a 2-lane reference dive). Key finding: HL2 CW is **100 % gateway-keyed** — no host DSP/WDSP carrier synthesis, EP2 TX-I/Q bypassed during CW; the host sends key-state bits (the verbatim packer is ALREADY in tree at NetworkProto1.cpp:99-108, dormant). Build order C-1..C-5 in the design doc; watch the **0x0b C&C; register collision** with the shipped ATT-on-TX step-attenuator before composing CW keyer config.

Other open major arcs (unchanged): **RX2 + Split** (#96–#101) and

PureSignal (Stage G/H). Smaller items: VOX (#91), voice keyer (#89), DSP buffer/latency options (#159), FM operator-knob refinement (#107), parked #156 (restart-after-hard-kill, intermittent).

*Last updated: 2026-06-17 EOD — **v0.4.0 RELEASED** (AM/DSB/FM transmit + AM carrier #106/#93, native TX DSP rack EQ/Speech/Combinator/Plate, Profiles

#49, TX monitor #90, ATT-on-TX §15.31, UI-readability batch). main is the trunk again: main==origin/main==tx-rebuild==v0.4.0. CW TX (#105) DESIGN MAPPED → docs/architecture/cw_tx_design.md. RX2/Split + PureSignal remain the next major arcs.*